

ALI AHMED KASSEM

Data Engineer, AI/ML Engineer

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PROFESSIONAL SUMMARY

Data Engineer and AI/ML Engineer with hands-on experience building ETL/ELT pipelines, relational data systems, cloud data workflows, and AI-powered automation solutions using Python, SQL, MySQL, Microsoft Azure, and n8n. Experienced in LLM integration, multimodal AI, workflow orchestration, and software architecture, with a focus on scalable, reliable, and production-oriented solutions.

PROFESSIONAL EXPERIENCE

Data Engineering Trainee, Microsoft Data Engineer Track, Digital Egypt Pioneers Initiative (DEPI)

Jul 2026 - Present, Hybrid, Egypt

- Selected for the Microsoft Data Engineer specialization track, developing practical skills in cloud data engineering and modern data platforms.
- Design and implement scalable cloud data workflows and big-data processing pipelines using Microsoft Azure technologies.
- Build end-to-end ETL/ELT pipelines, relational data models, and data management solutions using Python and MySQL.
- Apply automation and AI-assisted techniques to data engineering workflows, including deployment strategies and prompt engineering.

Artificial Intelligence Intern, Youth Technology Academy (YTA)

Dec 2024 - Jan 2025, Remote, Egypt

- Contributed to the AI development lifecycle through data preparation, model training, performance evaluation, and testing.

SELECTED PROJECTS

Lead Developer, Xare AI Platform

Mar 2026 - Present

- Designed and engineered Xare AI V2, a production-oriented multimodal AI platform combining AI services, workflow orchestration, and real-time application infrastructure.
- Built a 120-node n8n workflow orchestration system with intelligent model routing and multi-tier fallback across 5 AI model and speech services: Gemini, Qwen, GPT-OSS, Whisper, and Deepgram.
- Developed specialized pipelines for advanced reasoning, computer vision, document and PDF OCR, voice synthesis, and image generation.
- Developed a React/TypeScript frontend with real-time WebSocket voice streaming and IndexedDB-based media state management.
- Integrated cloud deployment infrastructure and reliability mechanisms to support modular, fault-tolerant AI workflows.

Team Lead and Software Developer, Supply Chain Management Simulation, University Project

Jan 2025 - Apr 2025

- Led the design and development of a comprehensive supply chain management simulation in Python, applying object-oriented programming and modular software architecture.
- Integrated a local LLM to automate business insights, customer review sentiment analysis, and marketing content generation.
- Engineered a desktop application with Tkinter and SQLite, implementing persistent data management and software design patterns including the Proxy pattern.

EDUCATION

Bachelor of Computer Science, E-JUST University, Expected June 2028

Specialization: Artificial Intelligence and Data Science

CERTIFICATIONS

AI for You: Training and Assessment, Oracle, Aug 2026

Introduction to Machine Learning Concepts, Microsoft, Dec 2025

Introduction to AI Concepts, Microsoft, Dec 2025

HCIA-AI V4.0 Course, Huawei, Nov 2025

TECHNICAL SKILLS

Data Engineering: Data Engineering, ETL/ELT Pipelines, SQL, Data Modeling, Relational Databases, Big Data Processing

Programming: Python, SQL, MySQL, JavaScript, TypeScript

Cloud & Data Platforms: Microsoft Azure, Firebase, SQLite, Cloud Data Workflows

AI & Machine Learning: Machine Learning, LLM Integration, Prompt Engineering, Multimodal AI, OCR, Image Classification

Automation & Systems: n8n Workflow Automation, Workflow Orchestration, Microservices, WebSockets, Fault-Tolerant Systems

Software & Tools: GitHub, Hugging Face, React, Tkinter, IndexedDB

LANGUAGES

Arabic — Native Proficiency
English — Professional Working Proficiency